

Module 2 – Arduino IDE Installation & Basic Electronics

Install, Connect & Make Your Arduino Come Alive!

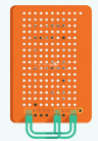




What Happens Today?



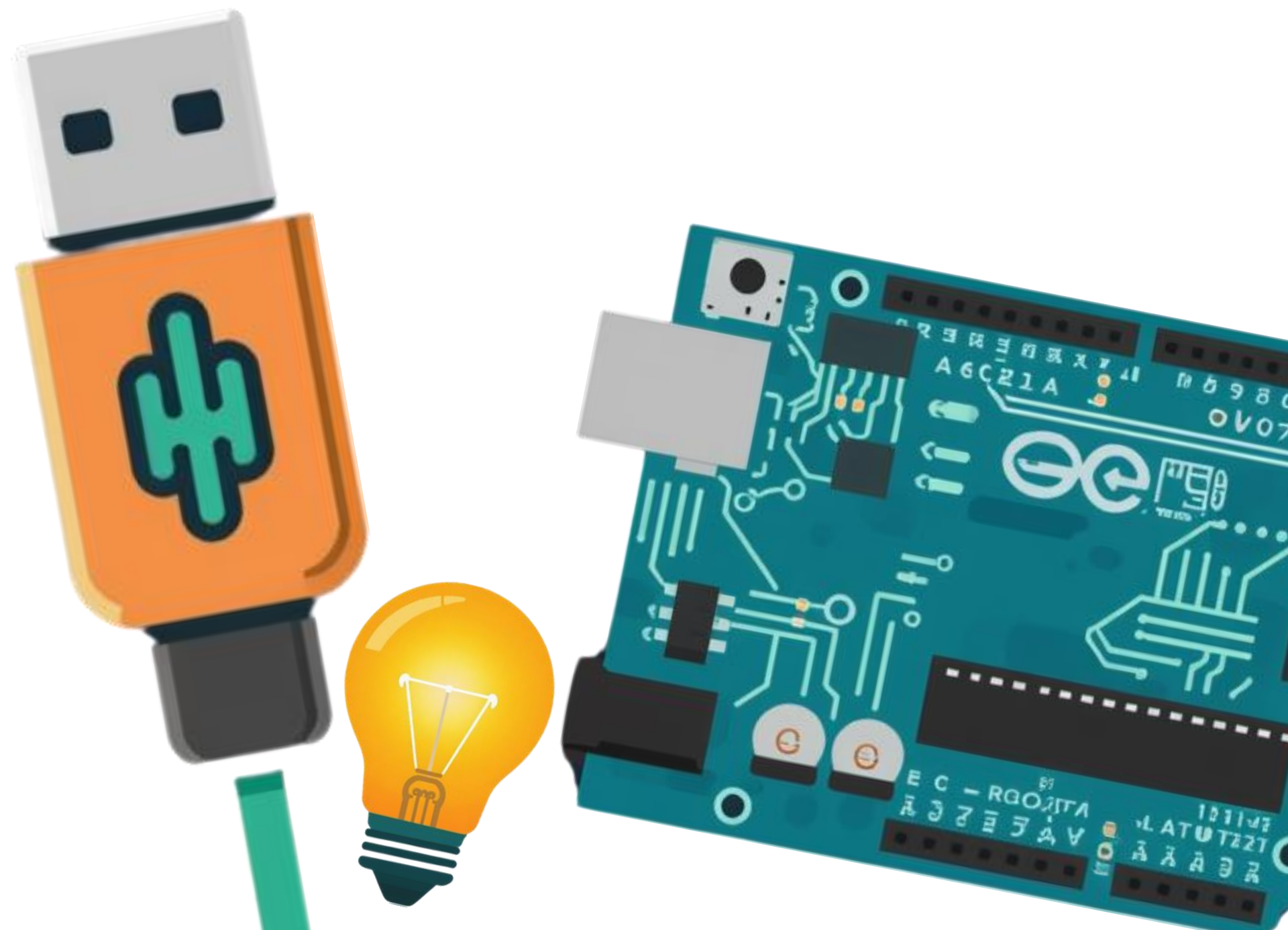
Install Arduino IDE & Connect your Arduino board via USB



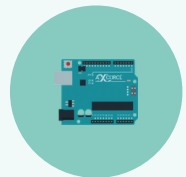
Learn breadboards & jumper wires, then identify key components



Upload your first code & troubleshoot common issues



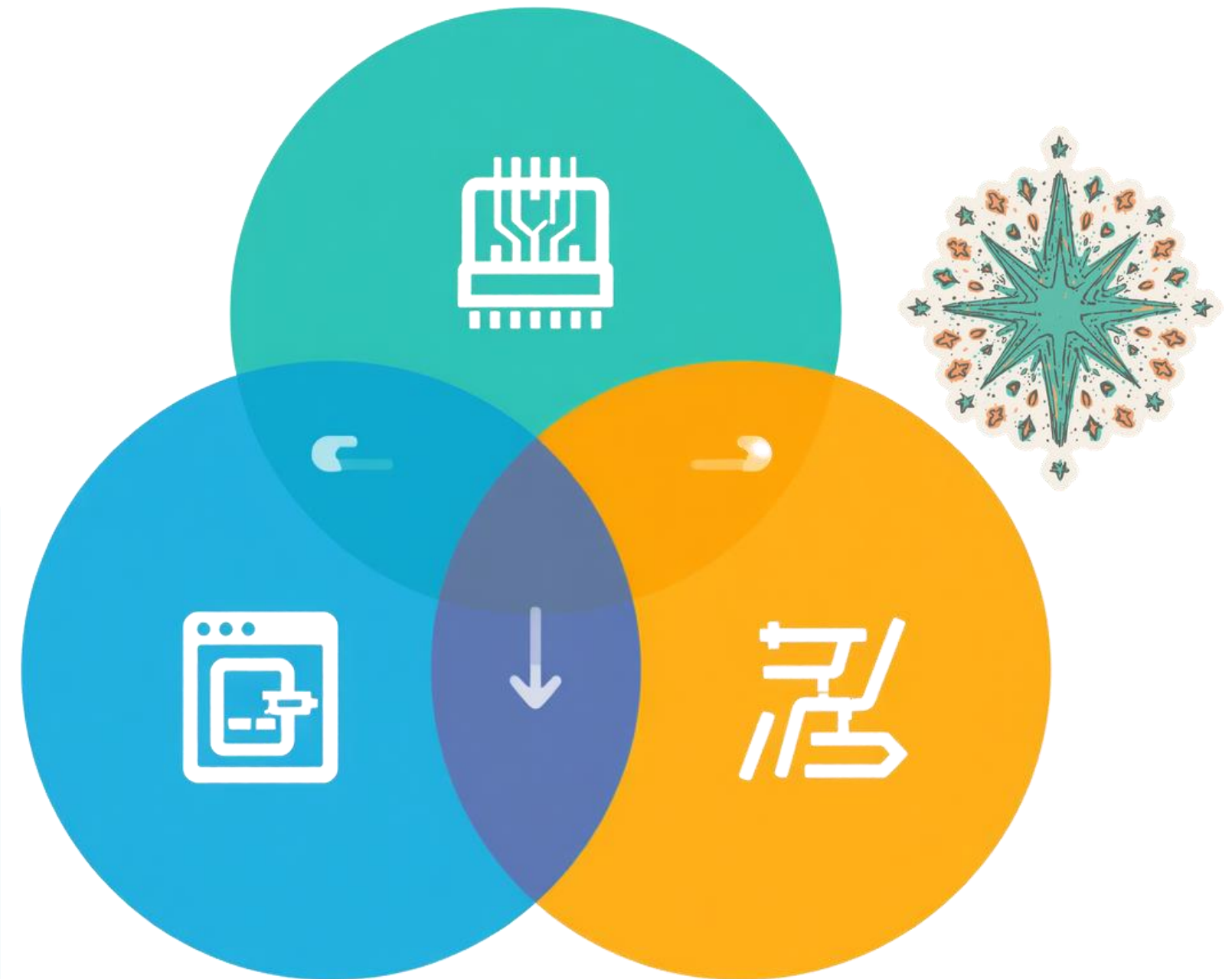
Meet the Arduino Ecosystem



Hardware + Software + Code work together — the Arduino board, the IDE, and your sketches form one connected system.



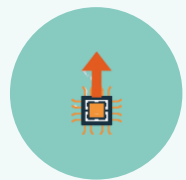
Each part depends on the others: the board runs your code, the IDE writes and uploads it, and code brings hardware to life.



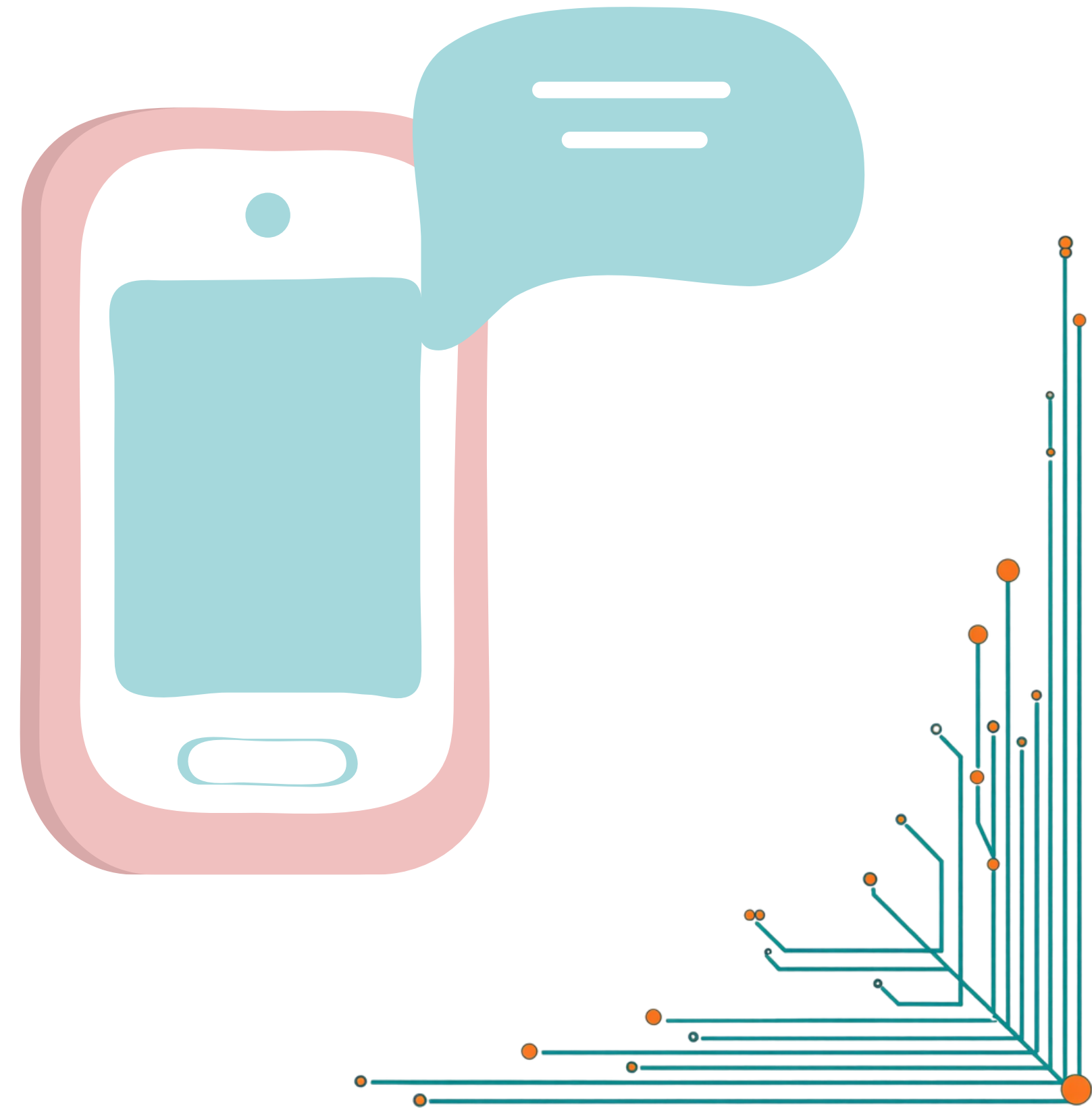
What is Arduino IDE?



Write, verify & upload your code here — the IDE is your coding command center for Arduino.



Highlights: code editor window, Verify button to check for errors, and Upload button to send code to your board.



Installing Arduino IDE



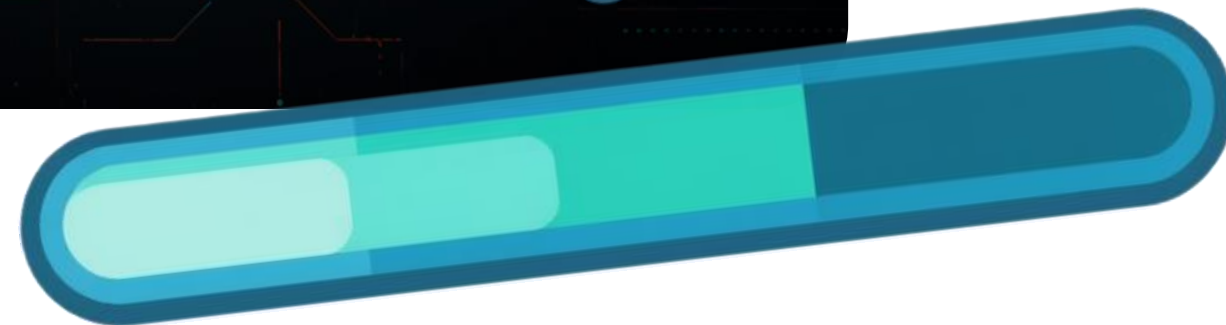
Step 1: Visit arduino.cc/en/software and download the IDE for your operating system.



Step 2: Run the installer and follow the on-screen prompts to complete setup.



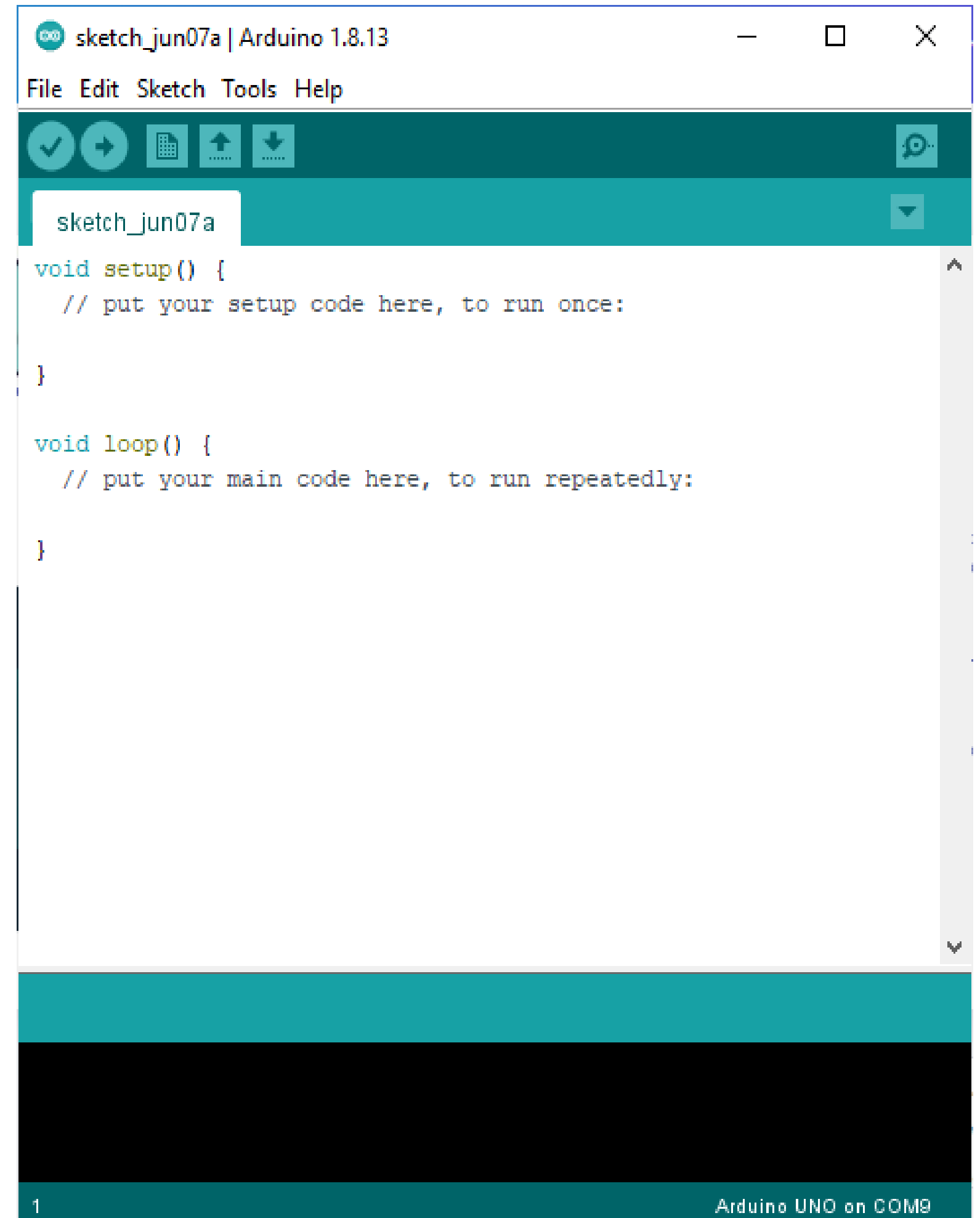
Step 3: Launch the Arduino IDE — you're ready to start coding your first project!





Arduino IDE Tour

- ✓ Port Selector & Board Selector — pick your USB port and choose your Arduino model.
- ✓ Code Editor — write your sketches here.
- ✓ Verify & Upload buttons — compile and send code to your board.
- ✓ Serial Monitor — view output



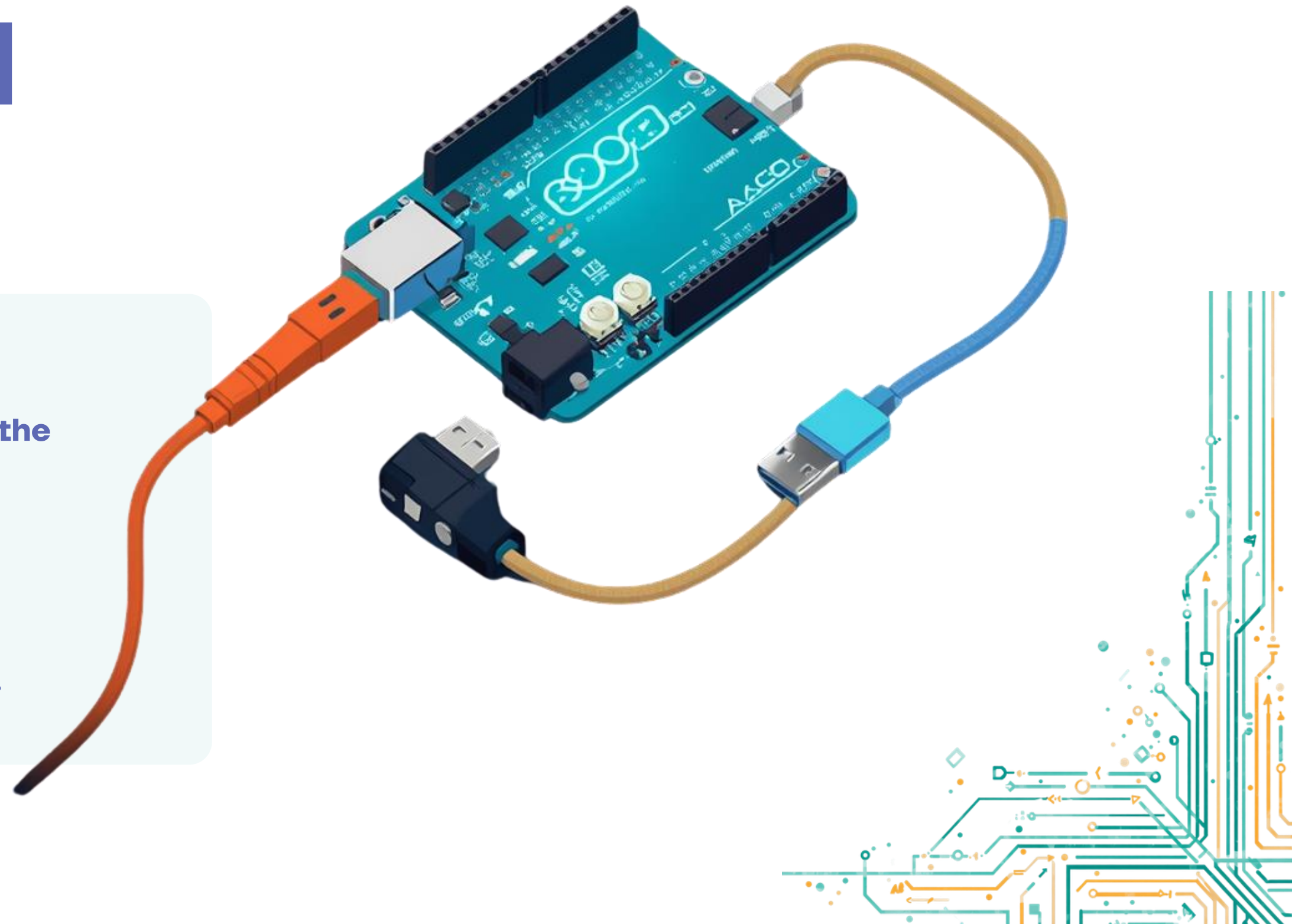
Connecting the Arduino Board



USB connection powers and links your board to your computer — plug in and the power LED lights up instantly!



Your computer will detect the Arduino automatically — check the port in Arduino IDE to confirm the connection.



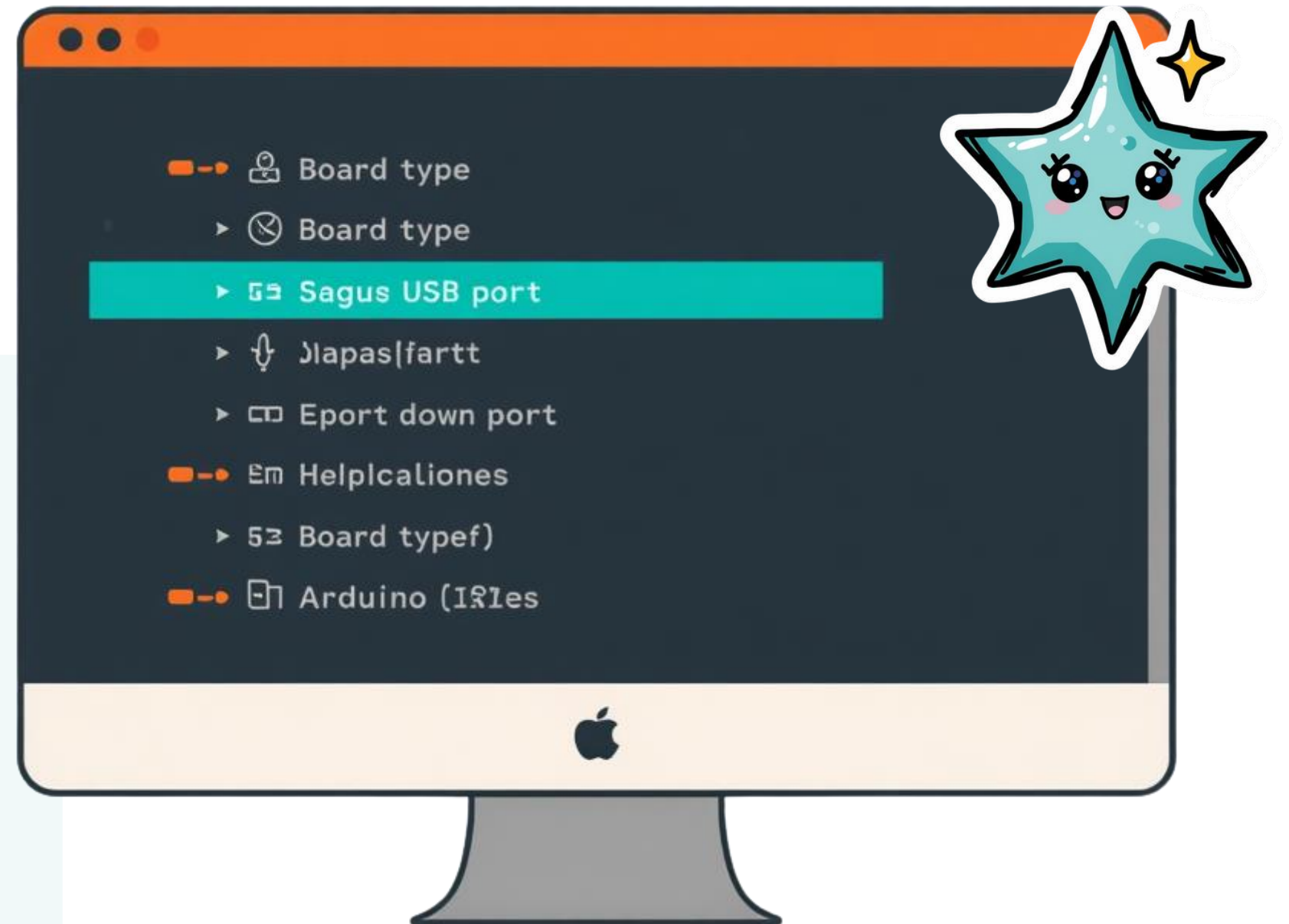
Board & Port Selection



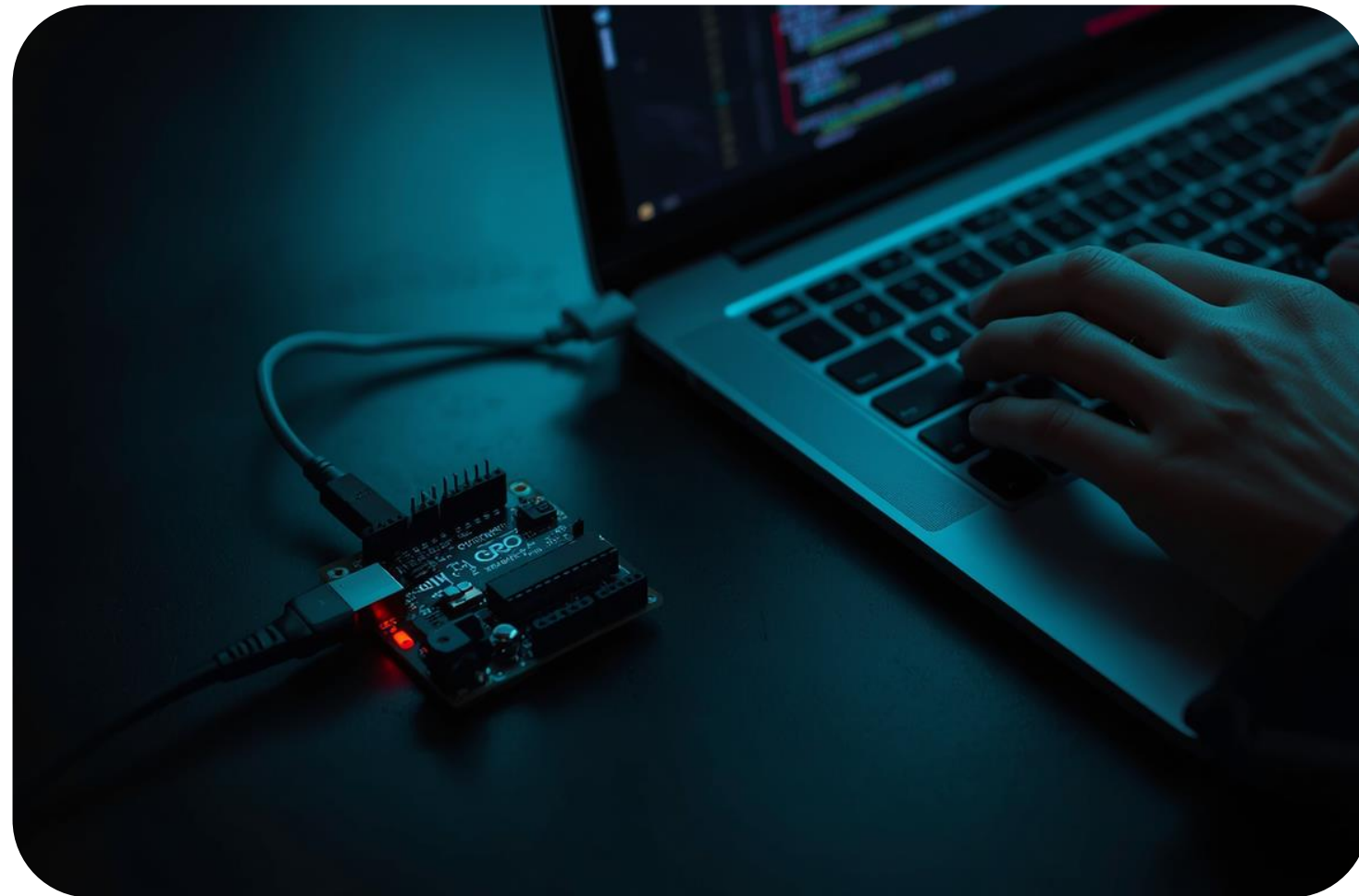
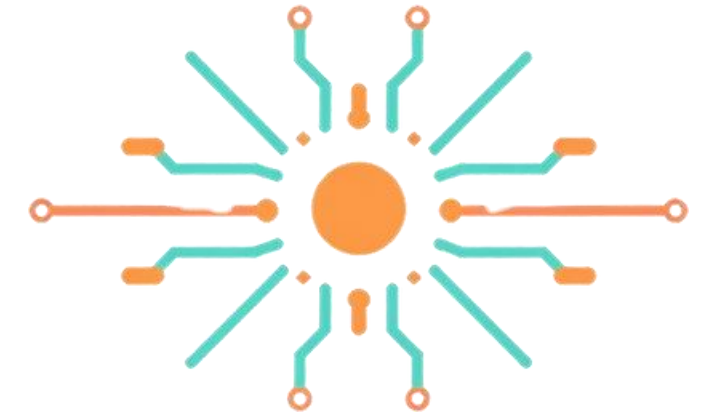
Select your board type from the Tools menu — choose Arduino Uno to match your hardware exactly.



Pick the correct USB port (COM on Windows) so your computer communicates with the board.



Understanding the Blink Workflow



Write your code in the Arduino IDE editor window.



Verify to check for syntax errors before uploading.



Upload to Arduino — watch the LED blink on the board!

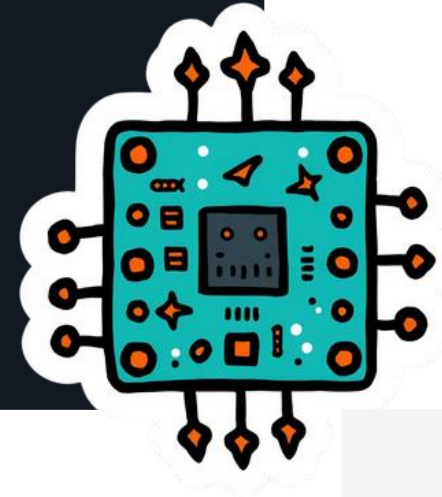
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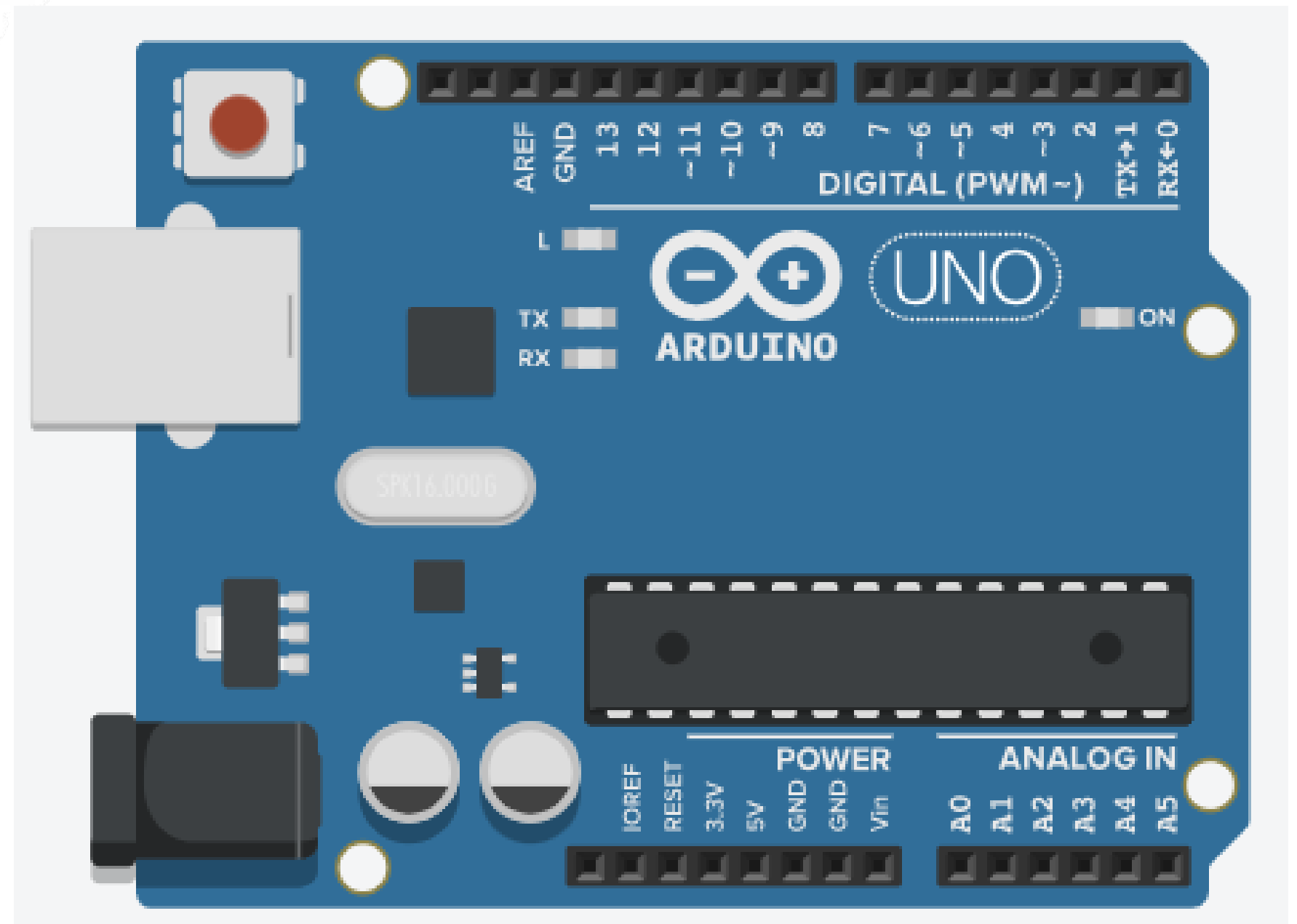
```



Make the inbuilt LED blink on and off — your very first Arduino sketch, and the "Hello World" of hardware programming!

First Arduino Program: Blink

Open a fresh blank sketch
 Open→0.1 Basics → Blink
 Upload
 Watch the inbuilt LED blink





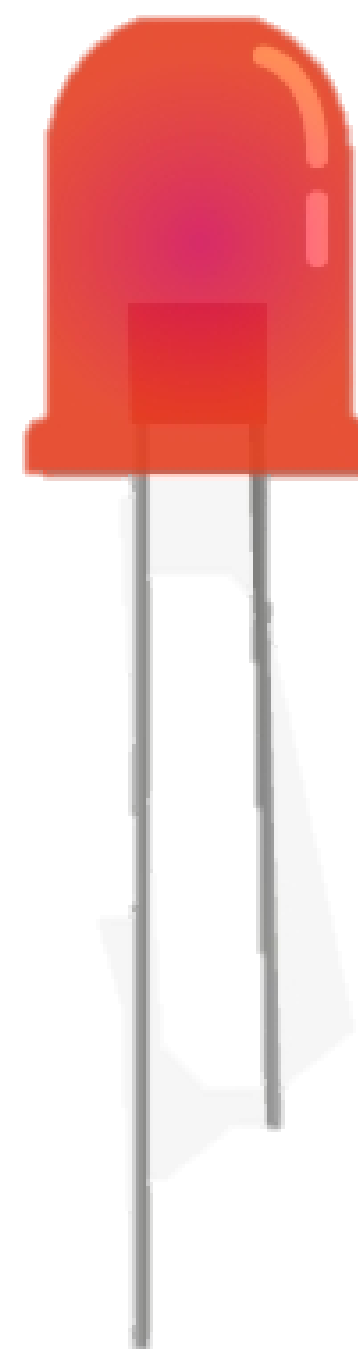
Component Spotlight: LED

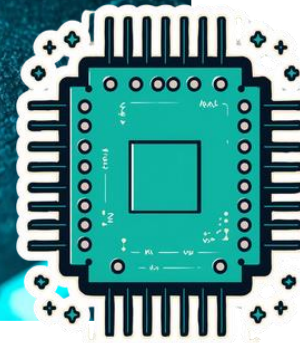
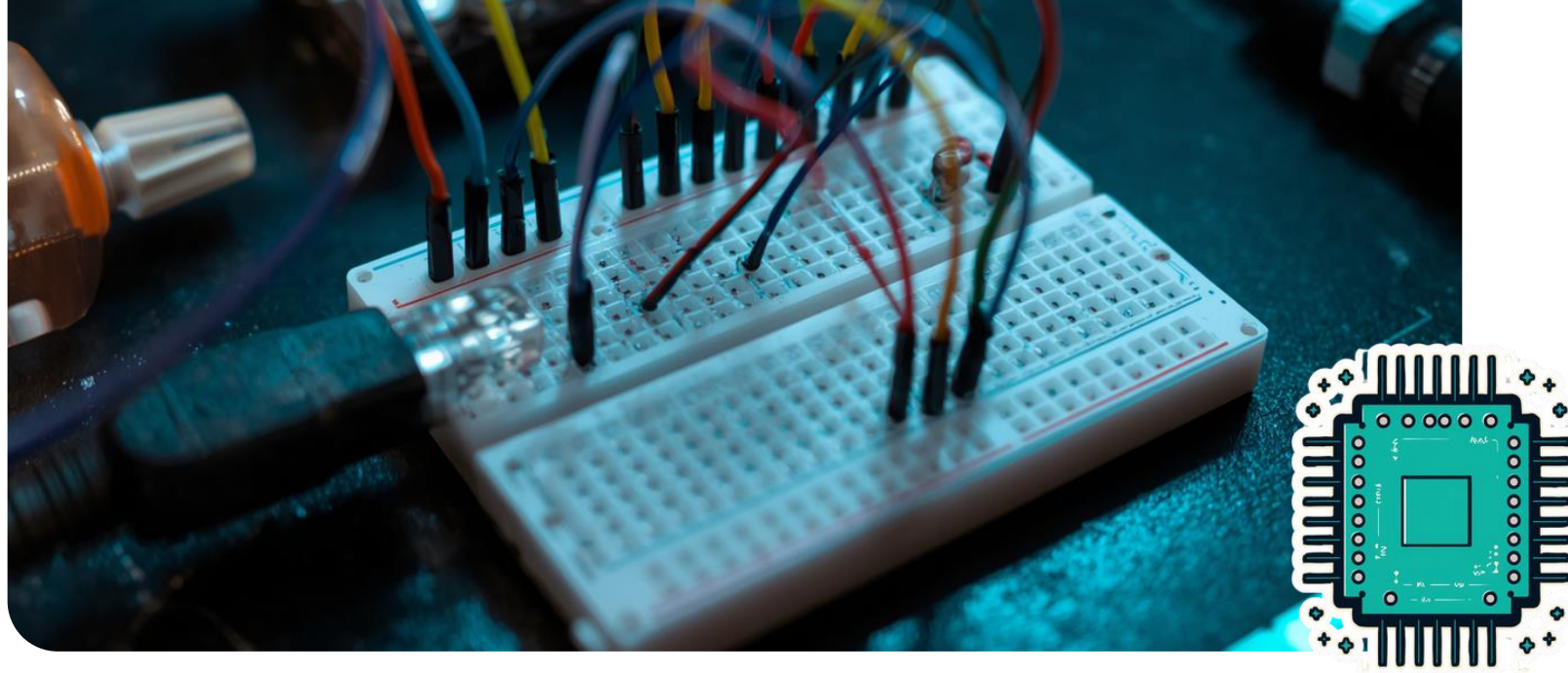


Long leg = Anode (+)
Connect to positive voltage. Current flows in through the anode to light up the LED.

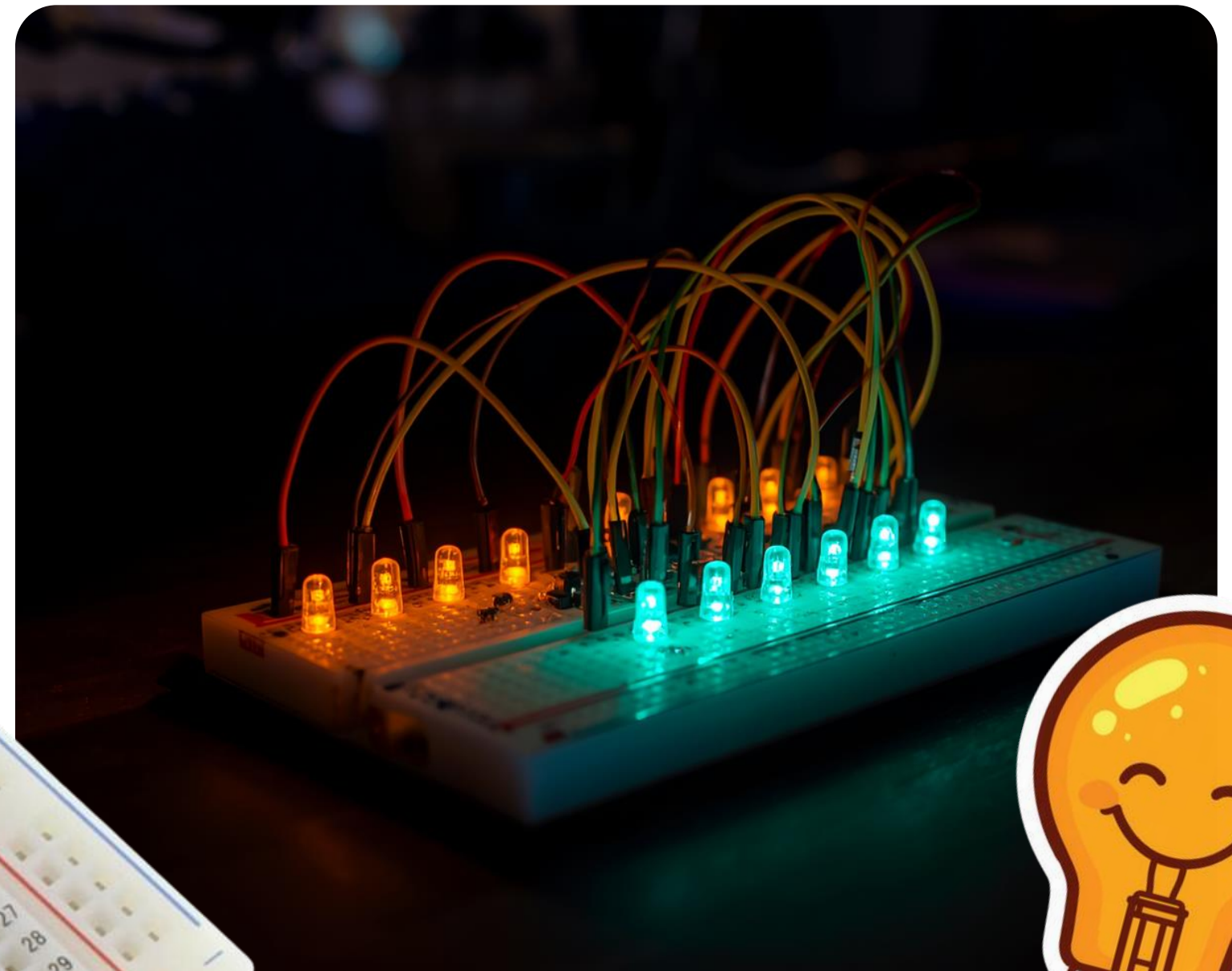
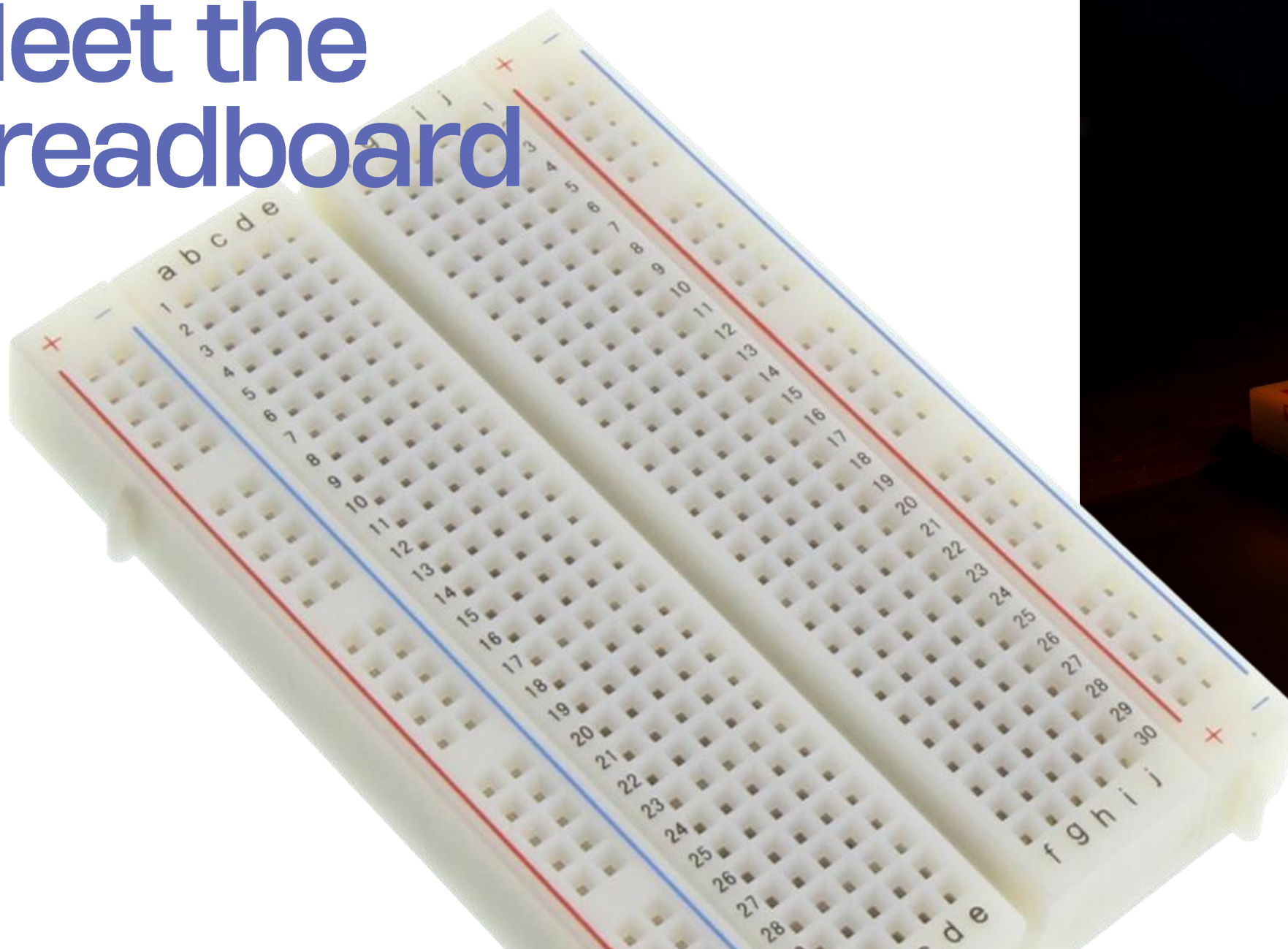


Short leg = Cathode (-)
Connect to ground. Current exits through the cathode to complete the circuit.

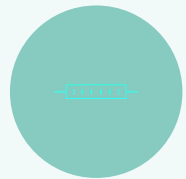




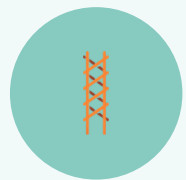
Meet the Breadboard



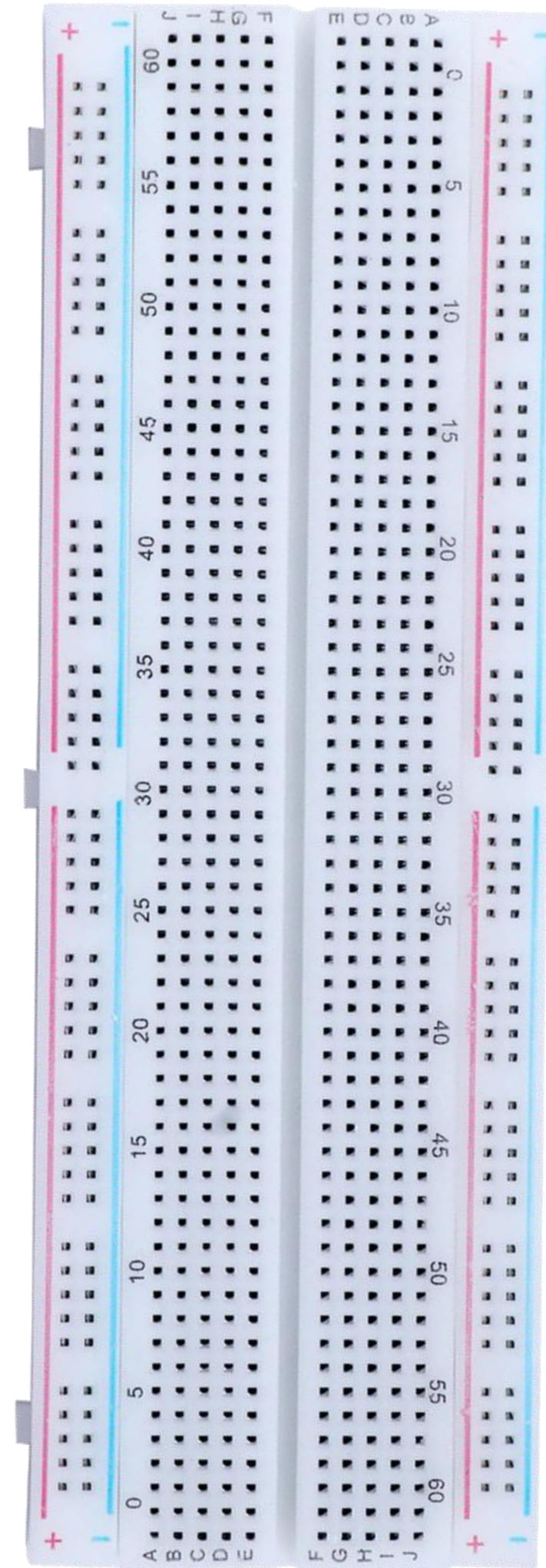
Inside a Breadboard



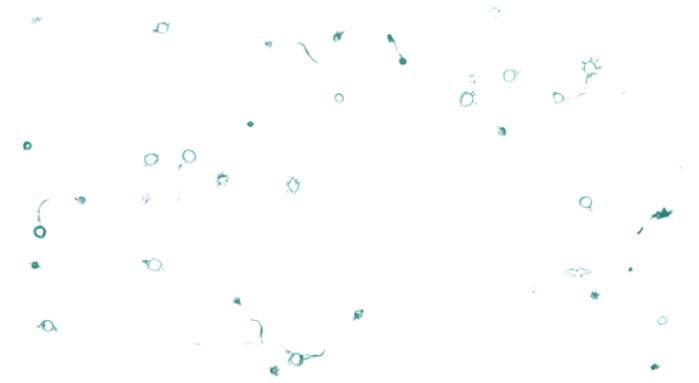
Metal strips run horizontally inside each row — holes in the same row share an electrical connection.



Power rails run vertically along the sides, connecting the entire column for easy + and – access.



And to connect... Jumper Wires



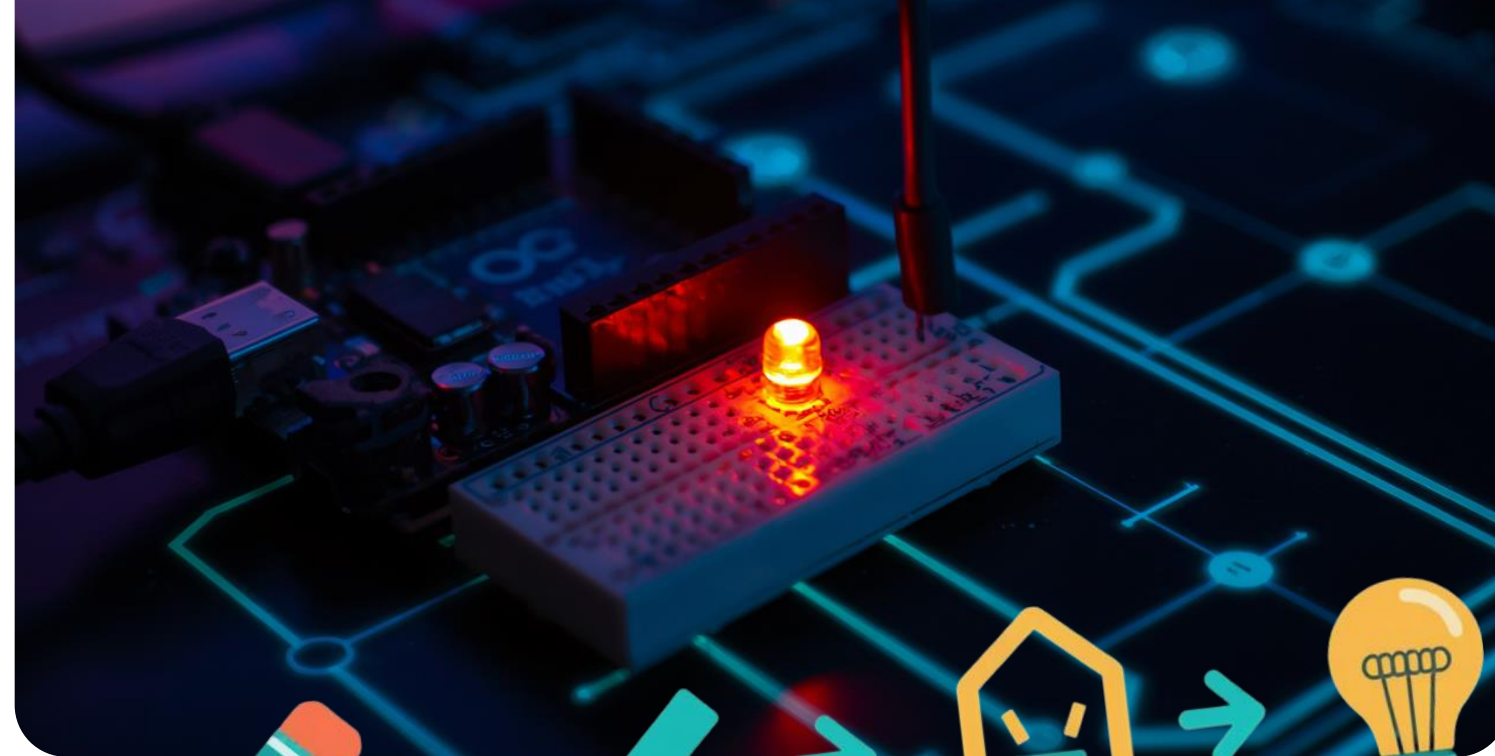
Male-Male: Connect two breadboard rows together easily.



Male-Female: Link Arduino pins to breadboard components.



Female-Female: Connect module pins to other modules.



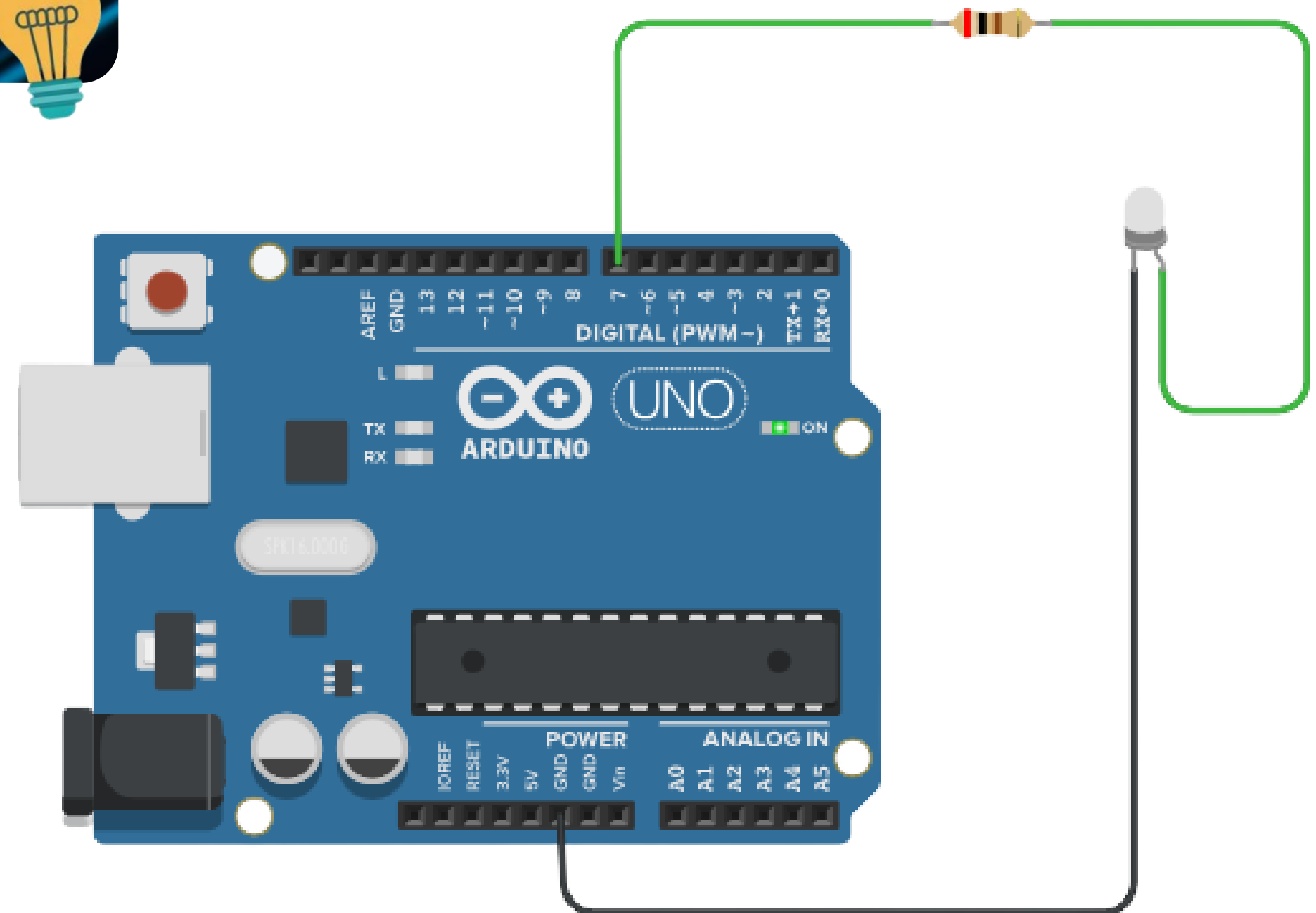
Make the LED blink on and off

Replace LED_BUILTIN with any number from 0 to 12

Connect your LED to that Digital pin through a 220/330Ω resistor (Use Breadboard)

If successful,
Declare a new variable:
`const int ledPin = 7;`

Replace pin number with the variable "ledPin"



Common Errors Beginners Make



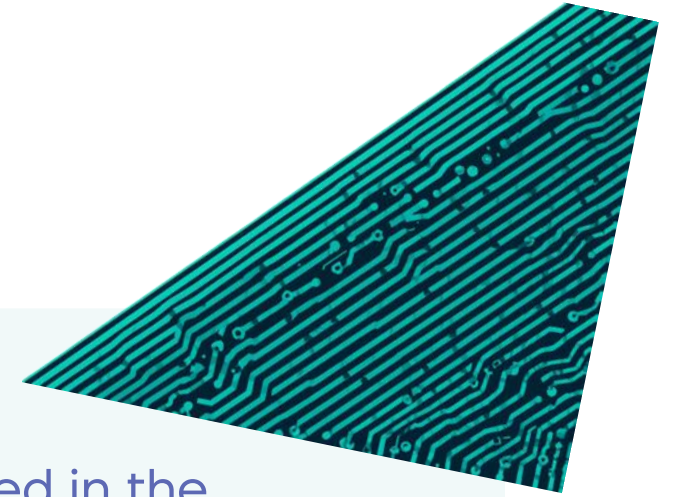
Wrong port or board selected in the IDE settings.



Loose USB connection or missing driver issues.



Syntax errors in code — missing semicolons or brackets!



Play Around

Try changing the delay,
and see what happens



What Happens During Upload?



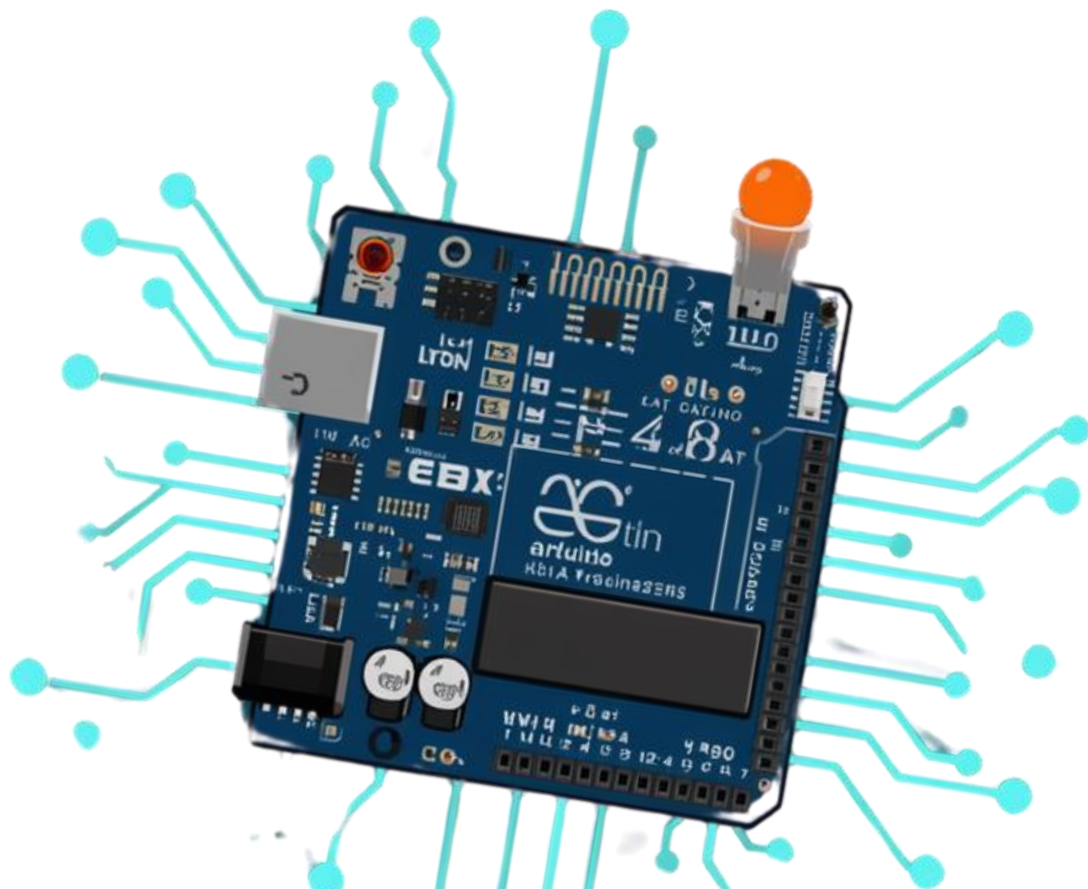
Your code travels from the computer through the USB cable directly into the Arduino's memory — ready to run instantly.



The Arduino stores your instructions in flash memory, so it remembers the program even when unplugged.



Arduino Success Checklist



IDE installed & board detected by your computer.



Correct port selected & Blink sketch uploaded successfully.



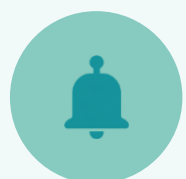
LED is blinking — your Arduino is alive!



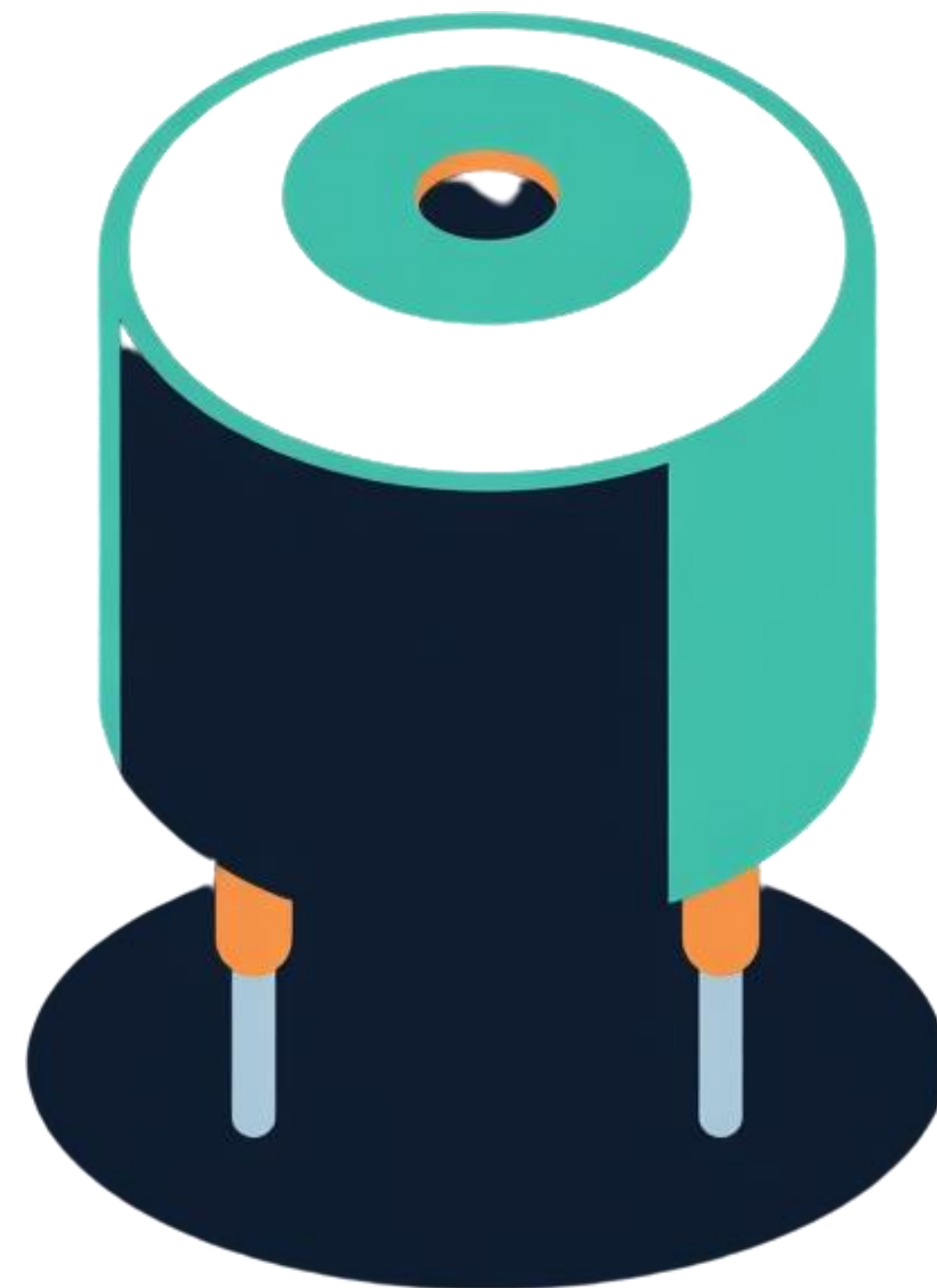
Component Spotlight: Buzzer



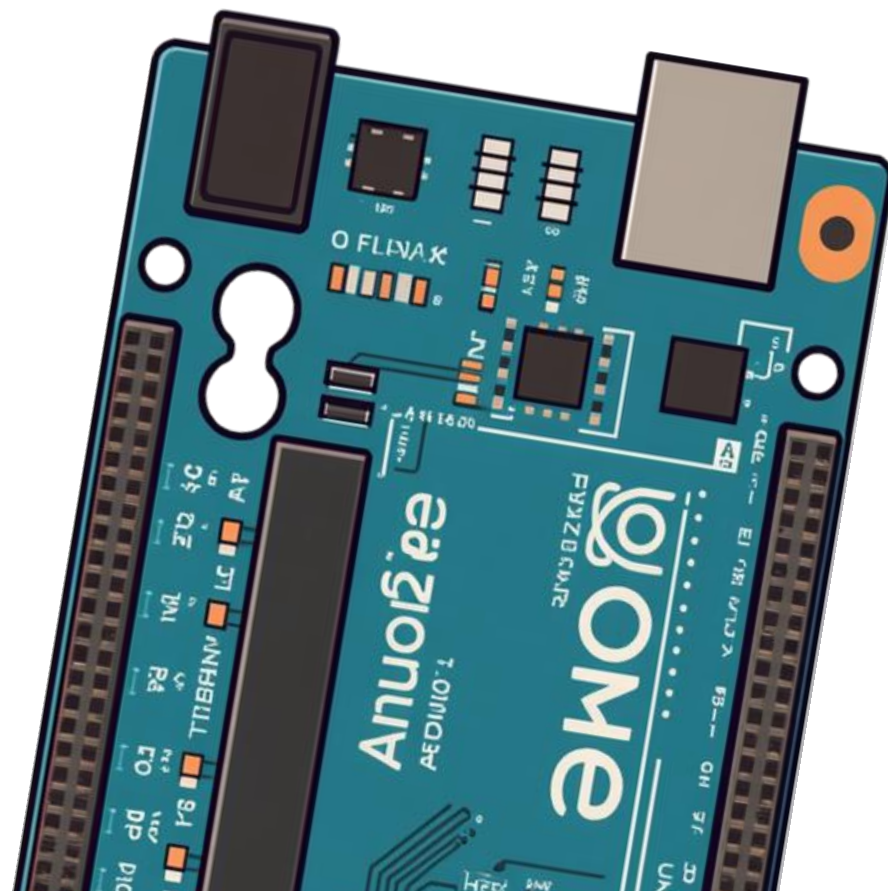
Produces sound when powered — used in alarms, alerts, and notifications



Connect to a digital pin and use `tone()` to control pitch and duration.



Module 2 Summary



Arduino IDE installed & board connected via USB



Breadboards, jumper wires & circuit basics mastered



Components identified & Blink program uploaded!

